Embedded Systems Programming

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COMMUNICATION INTERFACE

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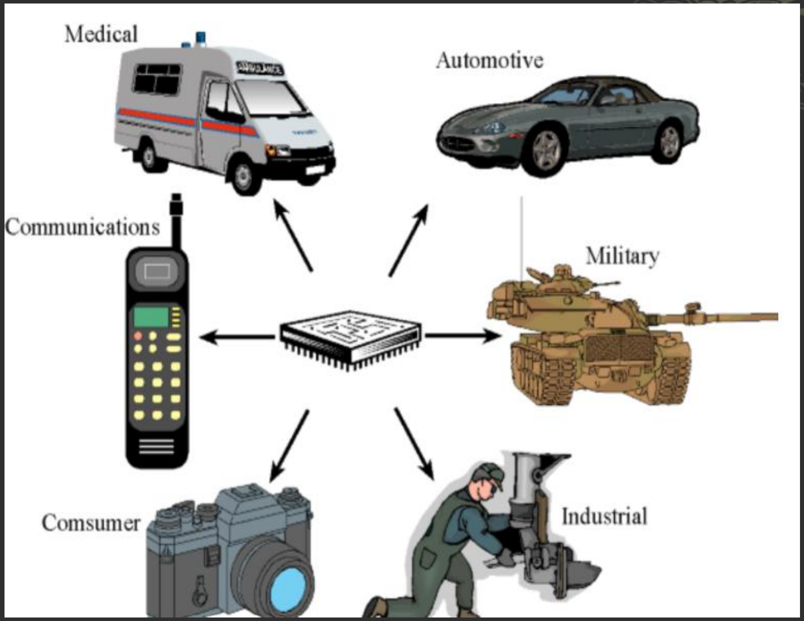
Overview

- When you implementing ES project you need to connect different devices such as amicrocontroller/ microprocessor to a sensor, display, or other module,….
  - Do you ever think about how the two devices talk to each other?
  - What exactly are they saying?
  - How are they able to understand each other?
- Communication between electronic devices is like communication between humans
  - Both sides need to speak the same language
  - In electronics, these languages are called communication protocols



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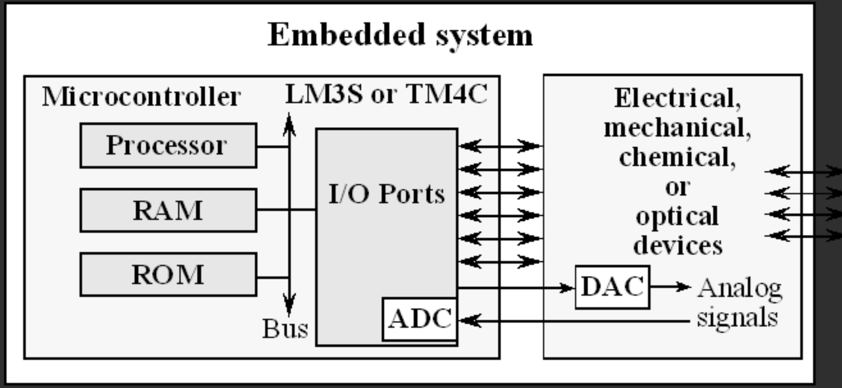
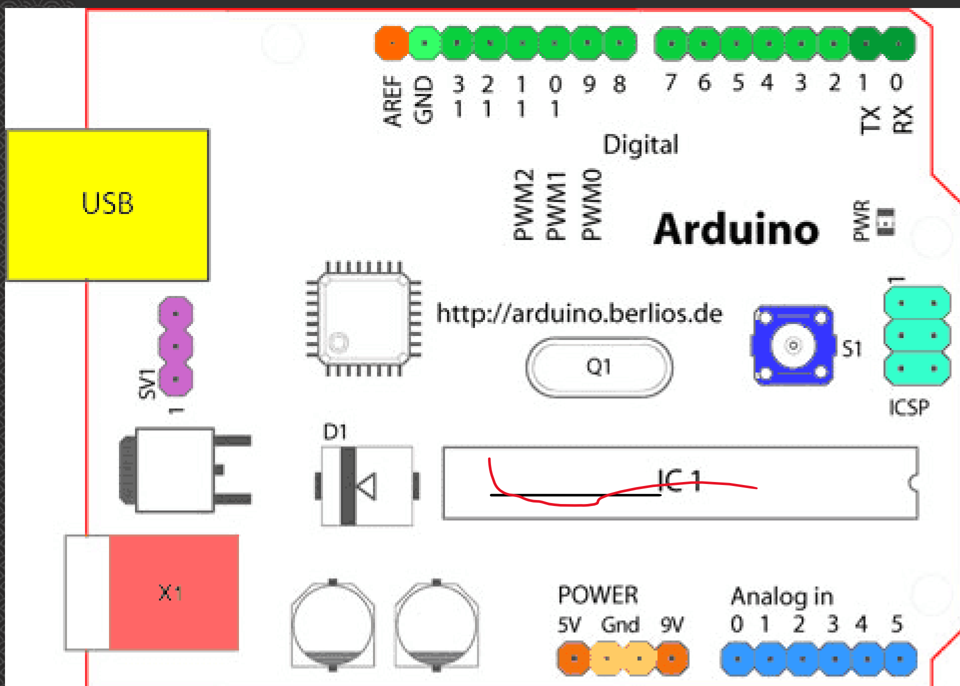
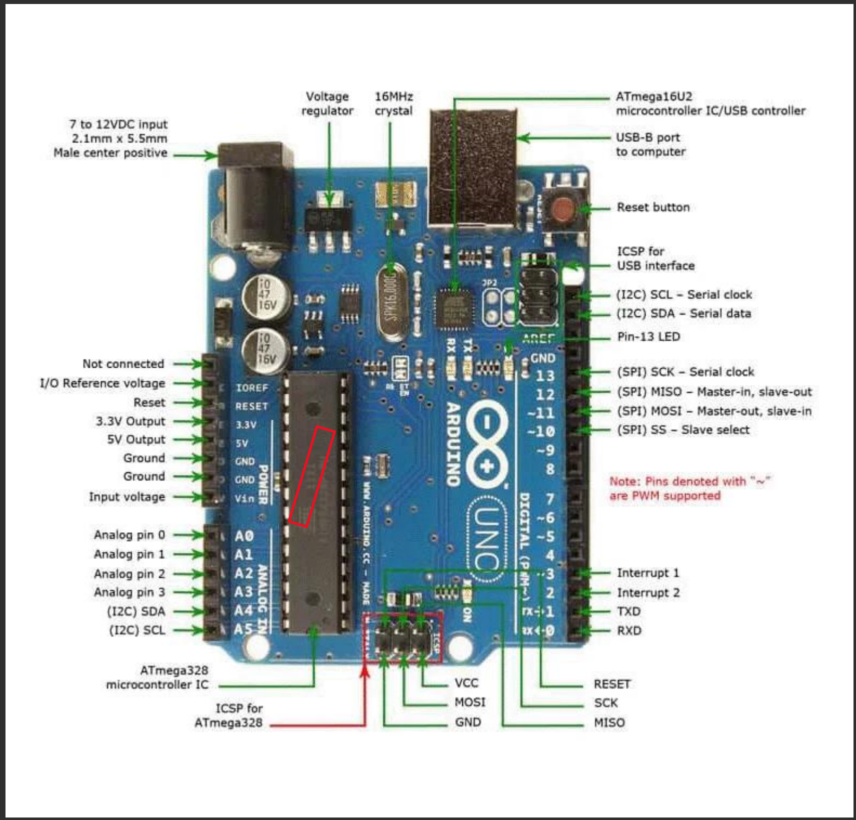
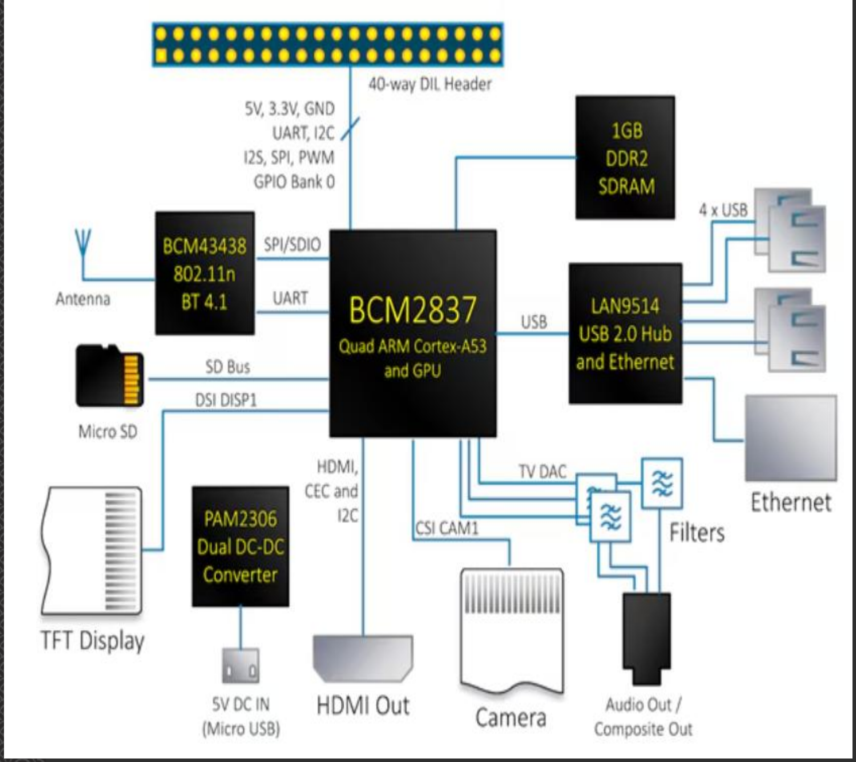


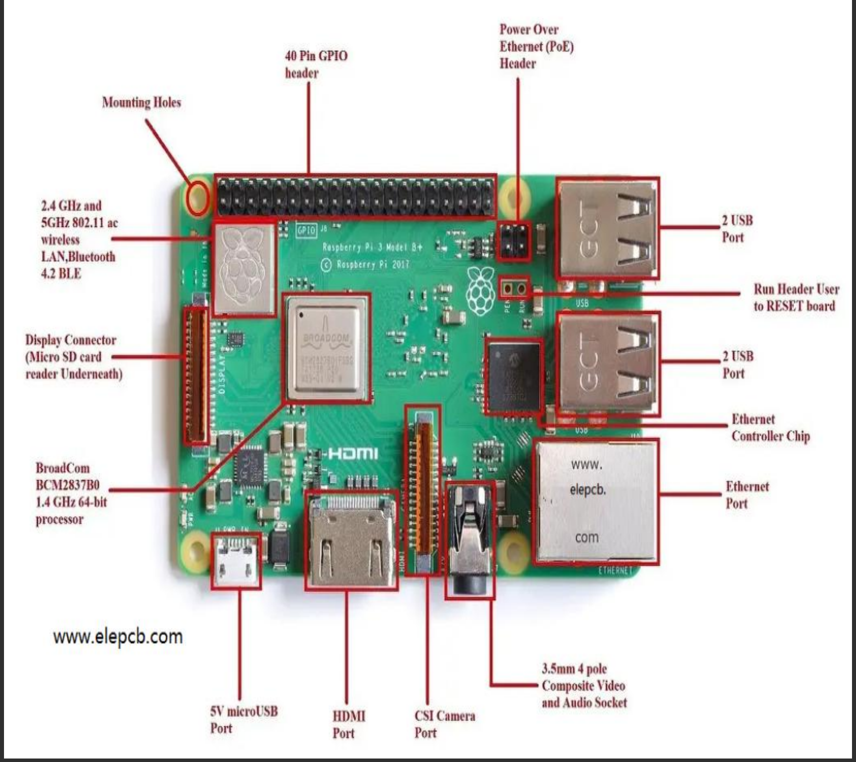
How are the Component communicating ?







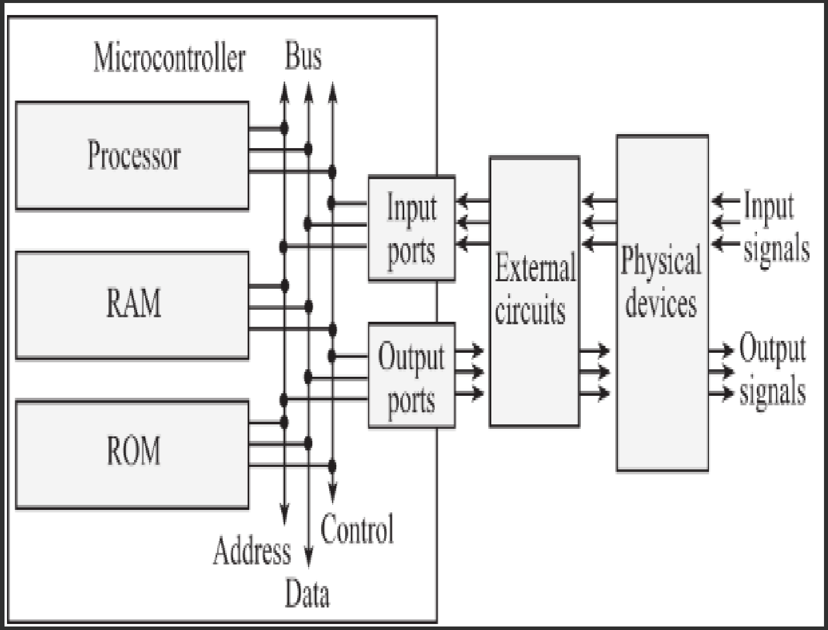
[Raspberry Diagram]



[Raspberry Pi]

How subsystem in ES are communicating ?

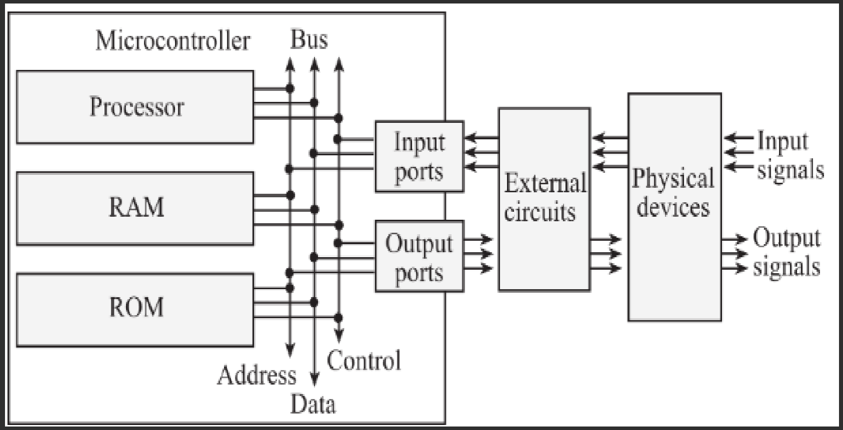
- A communication interface (CI) :
  - is essential for an ES to communicate with its subsystems or
    with external world
- There are two types of CIs for an ES:
  - Onboard CIs
  - and External CIs
- Embedded product is a combination of different types of components (chips/ devices) arranged on a printed circuit board (PCB)



An embedded system includes a microcomputer interfaced to external physical devices

How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Communication channels or buses used to connect various ICs and peripherals within the Esand are called Onboard Communication Interfaces
  - They are also called “Device/Board Level Communication Interfaces”.
- **Examples:**
  1. Inter-Integrated Circuit (I2C or I2C) Bus
  2. Serial Peripheral Interface (SPI) Bus
  3. UART
- 4.Controlled Area Network (CAN)
  4. Parallel Interface



An embedded system includes a microcomputer interfaced to external physical devices

How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Inter-Integrated Circuit (I2C or I2C) Bus
    - The Inter Integrated Circuit Bus (I2C–Pronounced 'I square C') is a synchronous bi-directional half duplex (one-directional communication at a given point of time) two wire serial interface bus
    - I2C bus was developed by 'Philips semiconductors' in the early 1980s
    - The original intention of I2C was to provide an easy way of connection between a microprocessor/ microcontroller system and the peripheral chips in television sets
    - The I2C bus comprise of two bus lines, namely; Serial Clock–SCL and Serial Data–SDA
    - SCL line is responsible for generating synchronisation clock pulses and SDA is responsible for transmitting the serial data across devices

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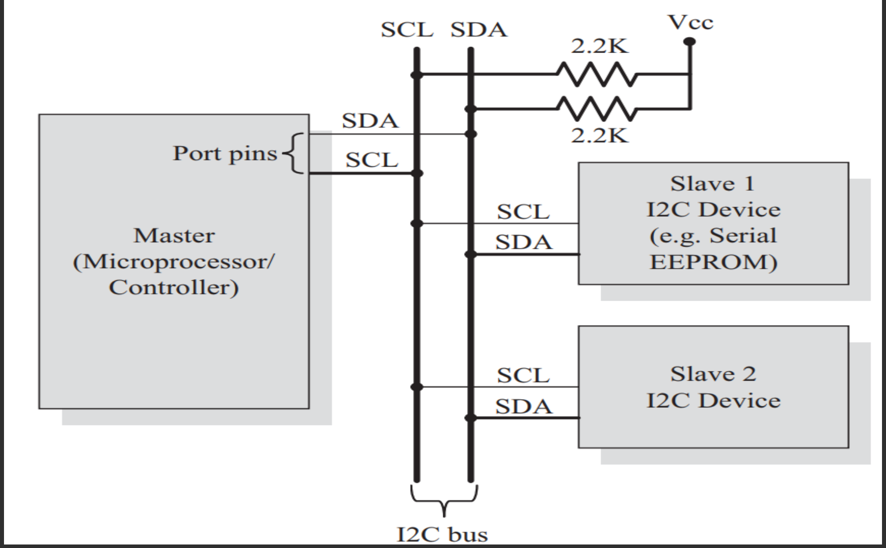
How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Inter-Integrated Circuit (I2C or I2C) Bus
    - I2C bus is a shared bus system to which many number of I2C devices can be connected
    - Devices connected to the I2C bus can act as either 'Master' device or 'Slave' device
    - The 'Master' device is responsible for controlling the communication by:
      - initiating/terminating data transfer,
      - sending data and
      - generating necessary synchronisation clock pulses



I2C Bus Interfacing

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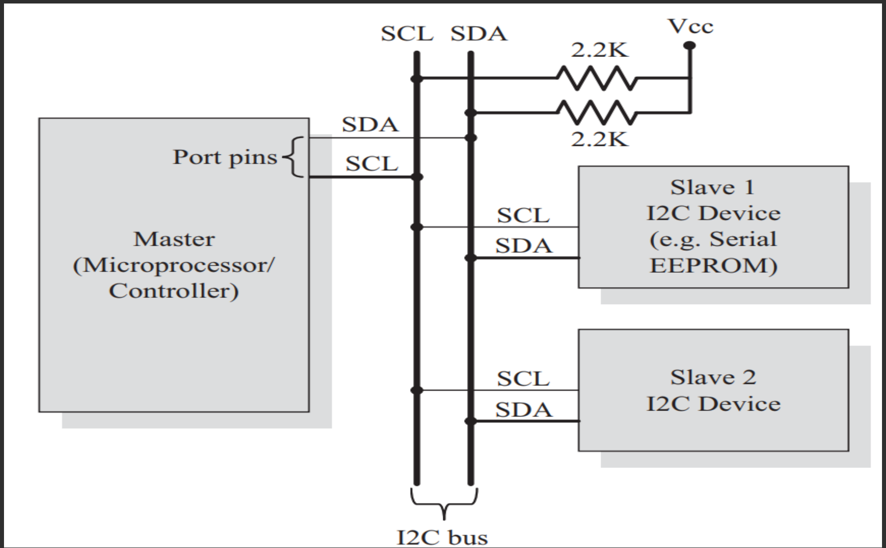
How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Inter-Integrated Circuit (I2C or I2C) Bus…..
    - ‘Slave’ devices wait for the commands from the master and respond upon receiving the commands
    - ‘Master’ and ‘Slave’ devices can act as either transmitter or receiver
    - Note that a master is acting as transmitter or receiver, the synchronisation clock signal is generated by the ‘Master’ device only
    - I2C supports multi masters on the same bus



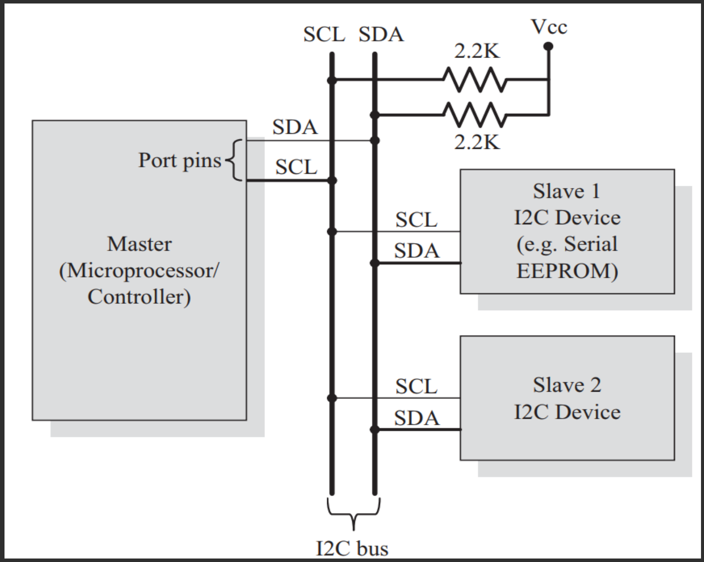
I2C Bus Interfacing

The sequence of operations for communicating with an I2C slave device is listed below

1. The master device pulls the clock line (SCL) of the bus to 'HIGH'
2. The master device pulls the data line (SDA) 'LOW', when the SCL line is at logic 'HIGH' (This is the 'Start' condition for data transfer)
3. The master device sends the address (7 bit or 10 bit wide) of the 'slave' device to which it wants to communicate, over the SDA line. Clock pulses are generated at the SCL line for synchronising the bit reception by the slave device. The MSB of the data is always transmitted fi rst. The data in the bus is valid during the 'HIGH' period of the clock signal
4. The master device sends the Read or Write bit (Bit value = 1 Read operation; Bit value = 0 Write operation) according to the requirement
5. The master device waits for the acknowledgement bit from the slave device whose address is sent on the bus along with the Read/Write operation command. Slave devices connected to the bus compares the address received with the address assigned to them

The sequence of operations for communicating with an I2C slave device is listed below…..

6. The slave device with the address requested by the master device responds by sending an acknowledge bit (Bit value = 1) over the SDA line
7. Upon receiving the acknowledge bit, the Master device sends the 8bit data to the slave device over SDA line, if the requested operation is 'Write to device'. If the requested operation is 'Read from device', the slave device sends data to the master over the SDA line
8. The master device waits for the acknowledgement bit from the device upon byte transfer complete for a write operation and sends an acknowledge bit to the Slave device for a read operation
9. The master device terminates the transfer by pulling the SDA line 'HIGH' when the clock line SCL is at logic 'HIGH' (Indicating the 'STOP' condition)



I2C Bus Interfacing

- **Data rates supported by I2C**
  - Standard mode: Data rate up to 100 Kbps
  - Fast Mode: up to 400 Kbps
  - High speed mode: up to 3.4 Mbps
- **Advantage**
  - Only uses two wires
  - Supports multiple masters and multiple slaves
  - ACK/NACK bit gives confirmation that each frame is transferred successfully
  - Hardware is less complicated than with UARTs
  - Well known and widely used protocol

- Disadvantages
  - Slower data transfer rate than SPI
  - The size of the data frame is limited to 8 bits
  - More complicated hardware needed to implement than SPI

Applications of I2C

- Reading some memory ICs
  - I2C is widely employed in memory modules such as EEPROM (Electrically Erasable Programmable Read-Only Memory),
  - FRAM (Ferroelectric Random-Access Memory),
  - and real-time clocks.
  - These modules utilize I2C for easy read-and-write operations, enabling data storage, timekeeping, and system configuration.
- Accessing DACs(digital-to-analog converters (DACs) and ADCs (analog-to-digital converters)
  - It allows the control of audio settings and parameters, such as volume control, audio routing, and digital signal processing (DSP) configurations
  - Audio Codecs
- Transmitting and controlling user-directed actions
- Communicating with multiple microcontrollers

- Reading hardware sensors
  - Many sensors, such as temperature sensors, humidity sensors, pressure sensors, accelerometers, gyroscopes, and magnetometers,
  - utilize I2C for easy integration with microcontrollers or embedded systems.
  - I2C allows efficient communication between the sensors and the control unit, enabling real-time data acquisition and monitoring
- Display Interfaces
  - I2C is commonly used in displays, including OLED (Organic Light-Emitting Diode) and LCD (Liquid Crystal Display) modules.
  - It allows the microcontroller or processor to control and configure the display parameters, such as brightness, contrast, and pixel data, using a simplified interface.
- Power Management
  - I2C is utilized in power management ICs (Integrated Circuits) for controlling and monitoring power-related functions in electronic systems.
  - It enables communication between power management ICs, voltage regulators, and other power control devices, facilitating power sequencing, voltage monitoring, and power supply management

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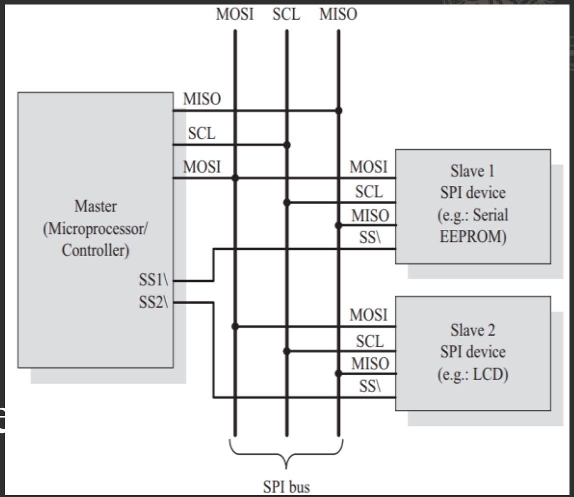
How are subsystem in ES are communicating ?

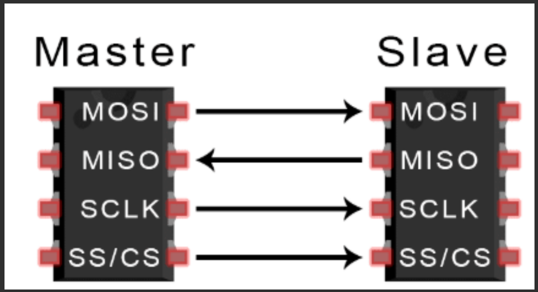
- **Onboard Communication Interfaces:**
  - Serial Peripheral Interface (SPI) Bus
    - a synchronous bi-directional full duplex, four-wire serial interface bus
    - SPI was introduced by Motorola.
    - One unique benefit of SPI is the fact that data can be transferred without interruption
    - Devices communicating via SPI are in a master-slave relationship
    - SPI is a single master multi-slave system
    - It is possible to have a system where more than one SPI device can be master, provided the condition only one master device is active at any given point of time, is satisfied
    - The master is the controlling device (usually a microcontroller), while the slave (usually a sensor, display, or memory chip) takes instruction from the master

How are subsystem in ES are communicating ?

- The simplest configuration of SPI is
  - a single master, single slave system, but one master can control more than one slave (more on this below)
  - SPI requires four signal lines for communication
    - **MOSI (Master Output/Slave Input)**
      - Line for the master to send data to the slave
      - known as Slave Input/Slave Data In (SI/SDI)
    - **MISO (Master Input/Slave Output)**
      - Line for the slave to send data to the master.
      - Known as Slave Output (SO/SDO)
    - **SCLK (Clock)**
      - Signal line carrying the clock signals
    - **SS/CS (Slave Select/Chip Select)** –
      - Signal line for slave device select.
      - It is an active low signal



SPI Bus Interfacing



SPI Bus Interfacing

The sequence of operations for communicating with an SPI slave device is listed below

1. The master device pulls the clock line (SCL) of the bus to ‘HIGH’
2. The master device pulls the data line (SDA) ‘LOW’, when the SCL line is at logic ‘HIGH’ (This is the ‘Start’ condition for data transfer)
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- **Data rates supported by SPI**
  - data at up to 60Mbps over short distances
  - SPI clock frequencies can reach up to 25MHz theoretically, but in practice, the maximum clock is limited to the frequency explicitly specified in the electrical specifications in the datasheet.
- **Advantage**
  - No start and stop bits, so the data can be streamed continuously without interruption
  - No complicated slave addressing system like I2C
  - Higher data transfer rate than I2C (almost twice as fast)
  - Separate MISO and MOSI lines, so data can be sent and received at the same time

- Disadvantages
  - Uses four wires (I2C and UARTs use two)
  - No acknowledgement that the data has been successfully received (I2C has this)
  - No form of error checking like the parity bit in UART
  - Only allows for a single master

Applications of SPI

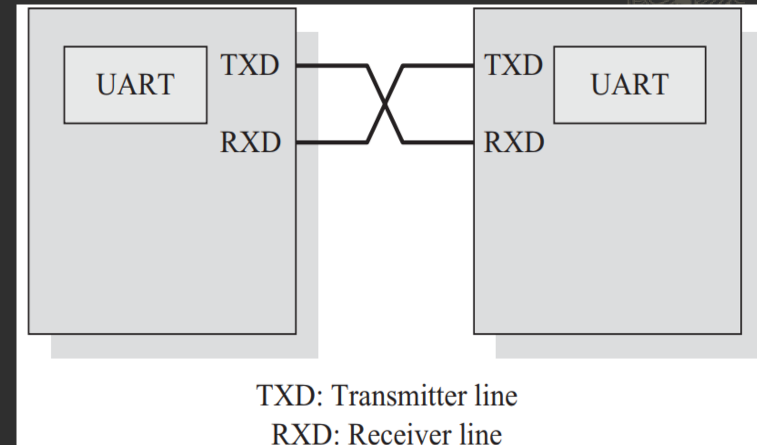
- For communicating between parameter measurement sensor and microcontrollers
- Memory: EEPROM, Flash, SD Card, MMC
- The SPI is used to talk to a variety of peripherals such as sensors i.e. temperature and pressure, analog to digital converter (ADC), digital to analog converter (DAC)
- Others: Liquide Crystal DisplayLCD, RTC(Real Time Clock), video game controller, Camera Lens Mount, touchscreen, etc
- Note That :
  - Compared to I2C, SPI bus is most suitable for applications requiring transfer of data in 'streams'.
  - The only limitation is SPI doesn't support an acknowledgement mechanism

How are subsystem in ES are communicating ?

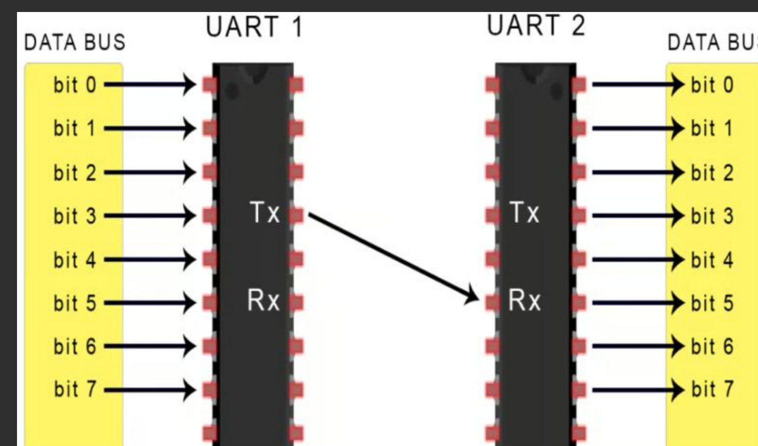
- **Onboard Communication Interfaces:**
  - Universal Asynchronous Receiver Transmiter(UART) Bus
    - (UART) based data transmission is an asynchronous form of serial data transmission
    - UART based serial data transmission doesn't require a clock signal to synchronise the transmitting end and receiving end for transmission
    - it relies upon the pre-defined agreement between the transmitting device and receiving device
    - The serial communication settings (Baudrate, number of bits per byte, parity, number of start bits and stop bit and flow control) for both transmitter and receiver should be set as identical
    - The start and stop of communication is indicated through inserting special bits in the data stream
    - To send a byte of data, a start bit is added first, and a stop bit is added at the end of the bit stream
    - The least significant bit of the data byte follows the 'start' bit
    - The 'start' bit informs the receiver that a data byte is about to arrive

How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Universal Asynchronous Receiver Transmiter(UART) Bus…….
    - The receiver device starts polling its ‘receive line’ as per the baudrate settings
    - If the baudrate is ‘x’ bits per second, the time slot available for one bit is 1/x seconds
    - The receiver unit polls the receiver line at exactly half of the time slot available for the bit
    - If parity is enabled for communication, the UART of the transmitting device adds a parity bit (bit value is 1for odd number of 1s in the transmitted bit stream and 0 for even number of 1s)



UART Interfacing



[Basics of UART Communication]

- **Data rates supported by** UART
  - Both the transmitter and receiver must be set to the same baud rate.
  - Mismatched baud rates can lead to data corruption.
  - Common baud rates include:
    - 300 bps
    - 1200 bps
    - 9600 bps
    - 115200 bps
- **Advantage**
  - Only uses two wires
  - No clock signal is necessary
  - Has a parity bit to allow for error checking
  - The structure of the data packet can be changed as long as both sides are set up for it
  - Low Cost: The simplicity of UART hardware makes it an economical choice for serial communication. It requires minimal components, making it suitable for budget-conscious projects
  - Ease of Use: With straightforward implementation and configuration, UART is accessible for both beginners and experienced engineers

Disadvantages

- The size of the data frame is limited to a maximum of 9 bits
- Doesn't support multiple slave or multiple master systems
- The baud rates of each UART must be within 10% of each other
- **Limited Distance:** UART is typically effective over short distances (up to a few meters)
- No Advanced Error Correction: While UART supports basic error detection through parity bits, it does not provide advanced error correction mechanisms, which might be necessary for critical applications

Applications of UART

- UART is widely used across various applications due to its simplicity and effectiveness:
  - Microcontroller Communication: Many microcontrollers, such as those from Arduino and Raspberry Pi, utilize UART for serial communication with peripherals like sensors, displays, and other microcontrollers
  - GPS Modules: UART is often used in GPS modules to transmit location data to microcontrollers, enabling navigation and mapping applications
  - Bluetooth and Wi-Fi Modules: Many wireless communication devices use UART to communicate with microcontrollers, facilitating data transfer over Bluetooth and Wi-Fi
  - Debugging and Console Interfaces: Developers often use UART for debugging embedded systems, allowing for easy output of diagnostic information during development

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How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces:**
  - Controlled Area Network (CAN)
    - CAN is a serial communication protocol used over a pair of wires
    - It is used in many realtime applications
    - It was developed by Robert Bosch to provide communication among various electronic components of vehicles
    - CAN specification doesn’t specify the layout and structure of the physical bus
    - A device connected to CAN bus is able to transmit on the physical bus
    - it was extended to automation and industrial applications
    - CAN protocol defines data packet format and transmission rules
      - To prioritize messages
      - Guarantee latency times
      - Handle transmission errors
      - Retransmit corrupted messages
      - Distinguish between a permanent failure of a node versus temporary errors

How are subsystem in ES are communicating ?

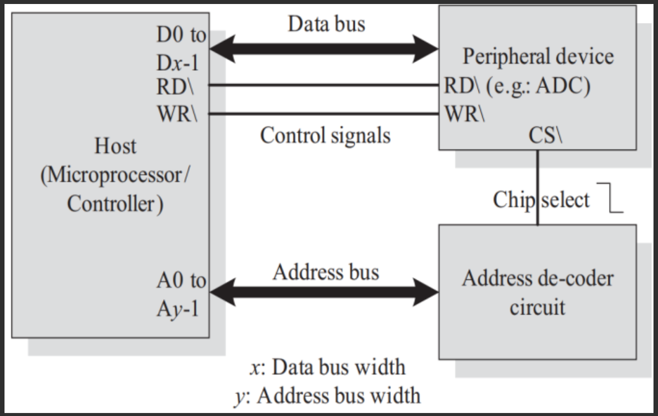
- **Important characteristics of CAN Protocol** :
  - Used in cars, buses, trucks and aircrafts in
    1. Safety systems like airbag control
    2. Antilock Brake System (ABS)
    3. Navigational systems like GPS
  - Real-time support
  - Offers medium speed (up to 125 Kbps) and high speed (up to 1 Mbps) data transfer
  - Error handling mechanism
- **Error correction in CAN:**
  - Error correction is typically done by a ‘retransmission and acknowledgement’ protocol.
  - Transmitter sends a data packet and expects to receive an acknowledgement from the receiver indicating that transmission is correct.
  - If an acknowledgement is not received in a specified time, the transmitter retransmits the data packet and waits for a second acknowledgement.

How are subsystem in ES are communicating ?

- **Applications of CAN**:
  - Used in cars, buses, trucks and aircrafts in
    1. Safety systems like airbag control
    2. Antilock Brake System (ABS)
    3. Navigational systems like GPS
  - Elevator Controllers
  - Photo Copiers
  - Medical instruments
  - Production line control systems

How are subsystem in ES are communicating ?

- **Onboard Communication Interfaces….**
  - **Parallel Interface**
    - In data transmission, parallel communication is a method of conveying multiple binary digits (bits) simultaneously, It contrasts with communication
    - The communication channel is the number of electrical conductors used at the physical layer to convey bits , Parallel communication implies more than one such conductor
    - For example,
      - an 8-bit parallel channel will convey eight bits (or a byte) simultaneously, whereas a serial channel would convey those same bits sequentially, one at a time.
      - Parallel communication is and always has been widely used within integrated circuits, in peripheral buses, and in memory devices such as RAM



Parallel Interface Bus

How are subsystem in ES are communicating ?

- **Important Features of Parallel Interface bus:**
  - Host processor has a parallel bus and control over read/write signals
  - Communication is controlled by ‘control signal interface’ between host and device
  - Read’ and ‘Write’ signals control the direction of data transfer It also provides control Signals
  - Each device connected to the processor is assigned a range of addresses
  - When the address selected is in this range, a decoder circuit activates the chip select line. Thus, the device becomes active
  - Processor initiates the parallel communication. If device wants to initiate communication, it sends an interrupt signal to the processor
  - Parallel data communication offers highest speed for data transfer
  - Width of the parallel interface can be 4-bit, 8-bit, 16-bit, 32-bit or 64-bits. It must match with the width of the data bus of the processor

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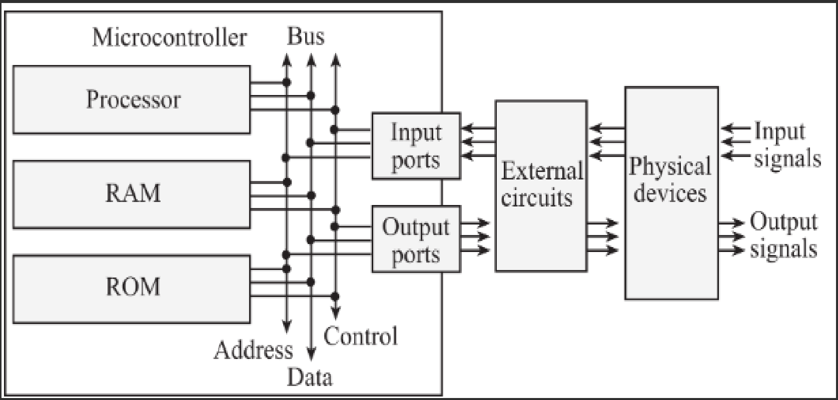
How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - They can be either wired media or wireless media and they can be a serial or parallel interfaces.
  - The External Communication Interface refers to the different communication channels/buses used by the embedded system to communicate with the external world
  - They are also called “Product Level Communication Interfaces”
  - **Examples:**

    1. RS-232C & RS-485
    2. Universal Serial Bus (USB)
    3. Infrared
    4. Bluetooth
    5. Wi-Fi
    6. ZigBee
    7. GSM
    8. General Packet Radio Service (GPRS)



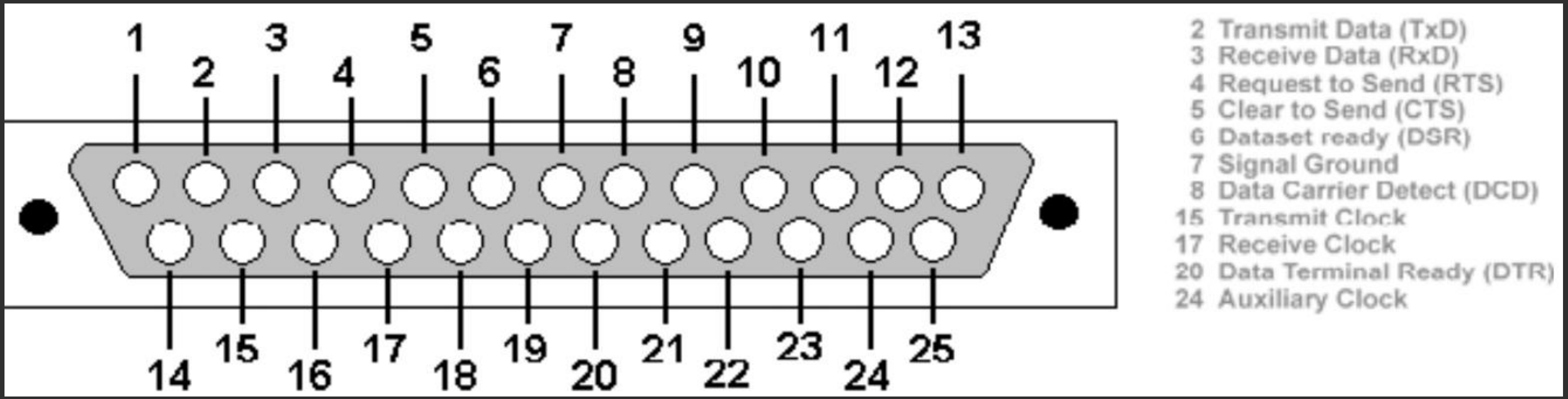
An embedded system includes a microcomputer interfaced to external physical devices

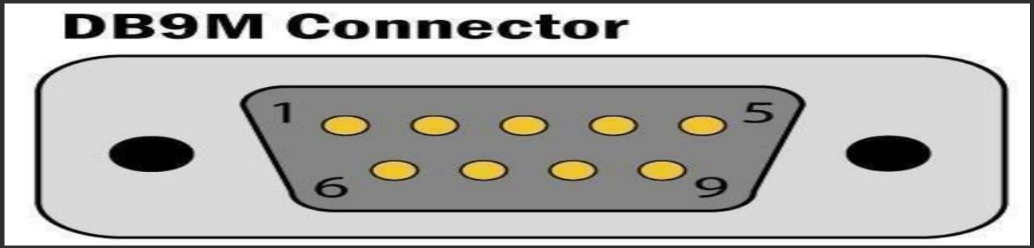
How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - RS-232C & RS-485
    - RS-232 C (Recommended Standard number 232, revision C from the Electronic Industry Association) is a legacy, full duplex, wired, asynchronous serial communication interface.
    - The RS-232 interface is developed by the Electronics Industries Association (EIA) during the early 1960s.
    - RS-232 extends the UART communication signals for external data communication
    - UART uses the standard TTL/CMOS logic (Logic „High" corresponds to bit value 1 and Logic „LOW" corresponds to bit value 0) for bit transmission whereas RS232 use the EIA standard for bit transmission
    - As per EIA standard, a logic „0" is represented with voltage between +3 and +25V and a logic „1" is represented with voltage between -3 and -25V
    - In EIA standard, logic „0" is known as „Space" and logic „1" as „Mark"

How the subsystem in ES ………… ?

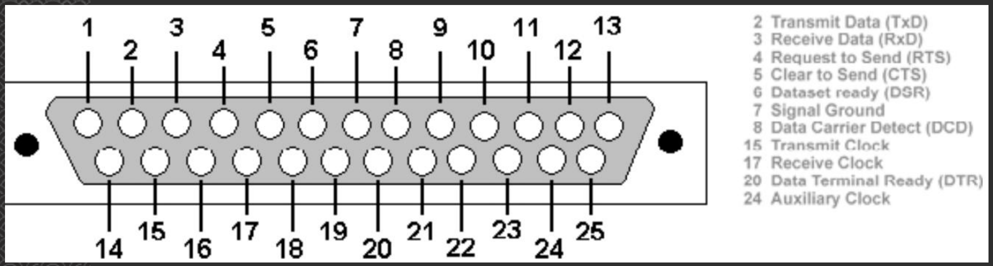
- **External Communication Interfaces:**
  - RS-232C & RS-485….
    - The RS232 interface define various handshaking and control signals for communication apart from the „Transmit" and „Receive" signal lines for data communication
    - RS-232 supports two different types of connectors, namely; DB-9: 9-Pin connector and DB-25: 25-Pin connector





How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - RS-232C & RS-485….
    - The RS232 interface define various handshaking and control signals for communication apart from the Transmit and Receive signal lines for data communication
    - RS-232 supports two different types of connectors, namely;
      - DB-9: 9-Pin connector and DB-25: 25-Pin connector
    - RS-232 is a point-to-point communication interface and the devices involved in RS-232 communication are called Data Terminal Equipment (DTE) and Data Communication Equipment (DCE)

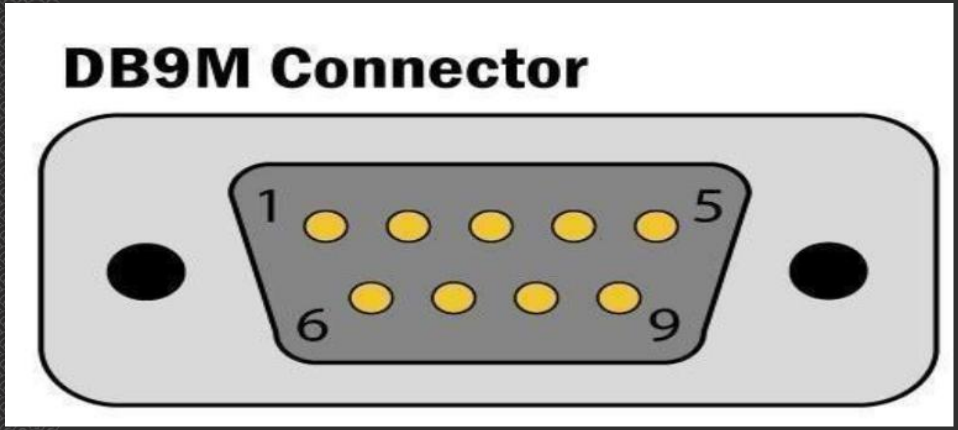


DB-25:25 Pin Connector

DB25 Male and Female Connector

How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - RS-232C & RS-485….
    - If no data flow control is required, only TXD and RXD signal lines and ground line (GND) are required for data transmission and reception
    - The RXD pin of DCE should be connected to the TXD pin of DTE and vice versa for proper data transmission
    - If hardware data flow control is required for serial transmission, various control signal lines of the RS-232 connection are used appropriately
    - The control signals are implemented mainly for modem communication and some of them may be irrelevant for other type of devices



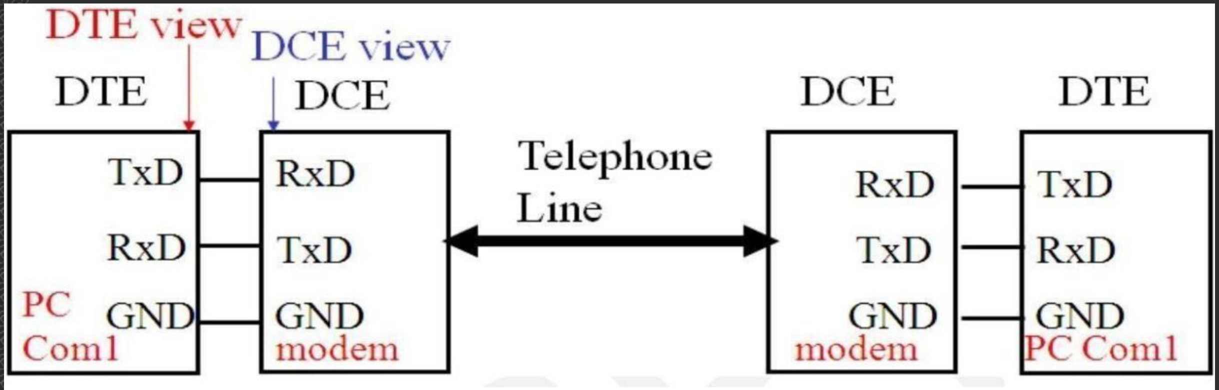
RS232 Pin Out

Pin #	Signal
1	DCD
2	RX
3	TX
4	DTR
5	GND
6	DSR
7	RTS
8	CTS
9	RI

- DB-9:9-Pin connector

How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - RS-232C & RS-485….
    - The Request to Send (RTS) and Clear To Send (CTS) signals co-ordinate the communication between DTE and DCE
    - Whenever the DTE has a data to send, it activates the RTS line and if the DCE is ready to accept the data, it activates the CTS line
    - The Data Terminal Ready (DTR) signal is activated by DTE when it is ready to accept data
    - The Data Set Ready (DSR) is activated by DCE when it is ready for establishing a communication link
    - DTR should be in the activated state before the activation of DSR
    - The Data Carrier Detect (DCD) is used by the DCE to indicate the DTE that a good signal is being received



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How the subsystem in ES ………… ?

- **External Communication Interfaces:**
  - RS-232C & RS-485….
    - Ring Indicator (RI) is a modem specific signal line for indicating an incoming call on the telephone line
    - As per the EIA standard RS-232 C supports baudrates up to 20Kbps (Upper limit 19.2Kbps)
    - The commonly used baudrates by devices are 300bps, 1200bps, 2400bps, 9600bps, 11.52Kbps and 19.2Kbps
    - The maximum operating distance supported in RS-232 communication is 50 feet at the highest supported baudrate
    - Embedded devices contain a UART for serial communication and they generate signal levels conforming to TTL/CMOS logic
    - A level translator IC like MAX 232 from Maxim Dallas semiconductor is used for converting the signal lines from the UART to RS-232 signal lines for communication
    - On the receiving side the received data is converted back to digital logic level by a converter IC
    - Converter chips contain converters for both transmitter and receiver
    - RS-232 uses single ended data transfer and supports only point-to-point communication and not suitable for multi-drop communication

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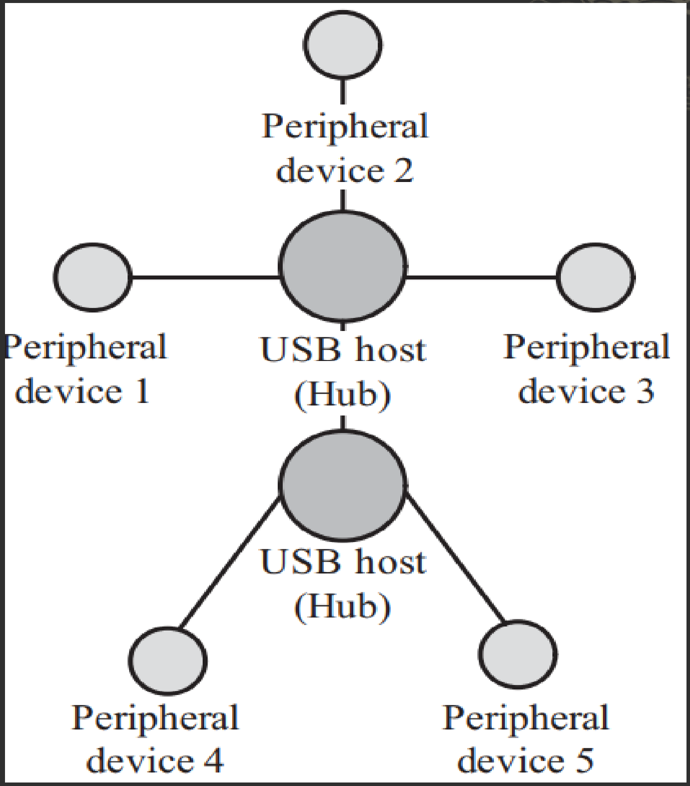
How the subsystem in ES ………… ?

- **External Communication Interfaces….:**
  - USB (UNIVERSAL SERIAL BUS)
    - USB is a wired high speed serial bus for data communication
    - The first version of USB (USB1.0) was released in 1995 and was created by the USB core group members consisting of Intel, Microsoft, IBM, Compaq, Digital and Northern Telecom
    - External Bus Standard
    - Allows connection of peripheral devices
    - Connects Devices such as keyboards, mice, scanners, printers, joysticks, audio devices, disk
    - Facilitates transfers of data at 480 (USB 2.0 only), 12 or 1.5 Mb/s (megabits/second)
    - Low-Speed: 10 – 100 kb/s
    - 1.5 Mb/s signaling bit rate
    - The USB cable in USB 2.0 specifi cation supports communication distance of up to 5 meters
    - Full-Speed: 500 kb/s – 10 Mb/s 12 Mb/s signaling bit rate
    - High-Speed: 400 Mb/s

How the subsystem in ES ………… ?

- **External Communication Interfaces….:**
  - USB (UNIVERSAL SERIAL BUS)…
    - The USB communication system follows a star topology with a USB host at the centre and one or more USB peripheral devices/USB hosts connected to it
    - USB transmits data in packet format. Each data packet has a standard format.
    - The USB communication is a host initiator and host contains a host controller which is responsible for controlling he data communication, including establishing connectivity with USB slave devices, packetizing and formatting the data packet
    - There are different standards for implementing the USB Host Control interface; namely Open Host Control Interface (OHCI) and Universal Host Control Interface (UHCI)
    - The physical connection between a USB peripheral device and master device is established with a USB cable



USB Device Connection topology

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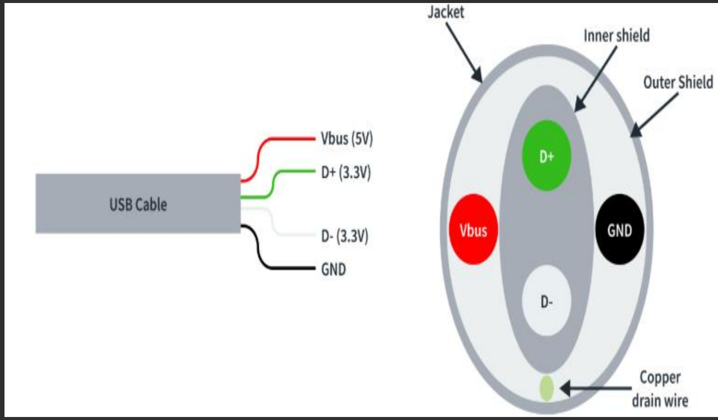
How the subsystem in ES ………… ?

- USB (UNIVERSAL SERIAL BUS)…
  - USB uses differential signals for data transmission. It improves the noise immunity
  - USB interface has the ability to supply power to the connecting device
  - Two connection lines (Ground and Power) of the USB interface are dedicated for carrying power
  - A Standard Downstream USB 2.0 Port (SDP) can supply power up to 500 mA at 5 V, whereas a Charging Downstream USB 2.0 Port (CDP) can supply power up to 1500 mA at 5 V
  - USB 3.2 (Generation1) is a super speed USB with 5Gbps of maximum data transfer speed
  - USB 3.2 (Generation2) is also a super speed USB with 10Gbps of maximum data transfer speed
  - This USB **4** is also known as Thunderbolt 4 with up to 40Gbps of maximum data transfer speed

Pin no.	Pin name	Description
1	V_{BUS}	Carries power (5V)
2	D–	Differential data carrier line
3	D+	Differential data carrier line
4	GND	Ground signal line



USB cable

How the subsystem in ES ………… ?

- Wireless communication interface
  - Wireless communication interface is an interface used to transmission of information over a distance without help of wires, cables or any other forms of electrical conductors
  - They are basically classified into following types not limited :
    - Infrared
    - Bluetooth
    - Wi-Fi
    - ZigBee
    - Lora
    - **Sigfox**
    - **GPRS**
    - GSM
    - Etc…

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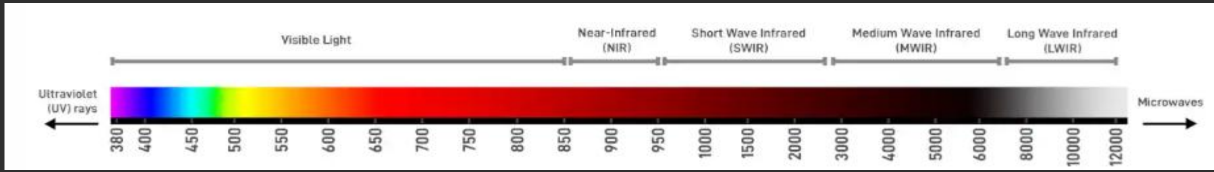
How the subsystem in ES ………… ?

- Wireless communication interface
  - Infrared(IR)
    - Infrared is a certain region in the light spectrum
    - Ranges from .7μ to 1000μ or .1mm
    - Broken into near, mid, and far infrared
    - One step up on the light spectrum from visible light
    - Measure of heat
    - Infrared is a serial, half duplex, line of sight based wireless technology for data communication between devices
    - It is in use from the olden days of communication for The remote control of your TV, VCD player etc
    - Works on Infrared data communication principle.
      - Infrared communication technique uses infrared waves of the electromagnetic spectrum for transmitting the data.
    - IR supports point-point and point-to-multipoint communication, provided all devices involved in the communication are within the line of sight



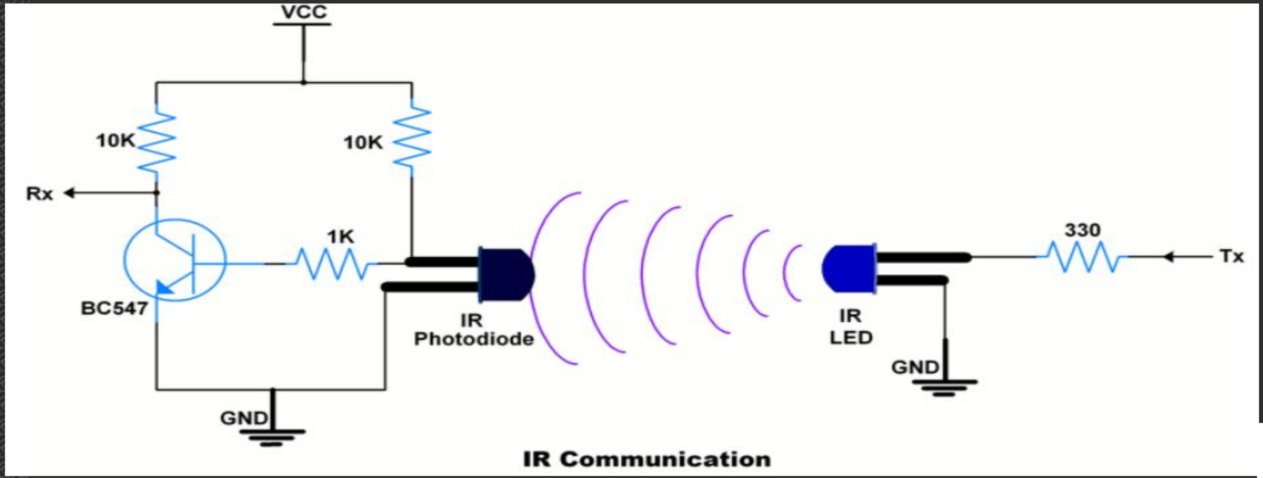
Visible & infrared(IR) Light Wavelength in nanometers(nm)

How the subsystem in ES ………… ?

- Wireless communication interface
  - Infrared(IR)……..
    - The typical communication range for IR lies in the range 10 cm to 1 m
    - The range can be increased by increasing the transmitting power of the IR device
    - IR supports data rates ranging from 9600bits/second to 16Mbps
    - Depending on the speed of data transmission IR is classified into
      - Serial IR (SIR) :Supports transmission rates ranging from 9600bps to 115.2kbps
      - Medium IR (MIR): Supports data rates of 0.576Mbps and 1.152Mbps
      - Fast IR (FIR): Supports data rates up to 4Mbps
      - Very Fast IR (VFIR):is designed to support high data rates up to 16Mbps
      - Ultra Fast IR (UFIR) :Supports data rates up-to 96Mbp
      - And GigaIR : Supports data rates 512 Mbps to 1 Gbps

How the subsystem in ES ………… ?

- Infrared(IR)……..
  - The typical communication range for IR lies in the range 10 cm to 1 m
  - IR communication involves a transmitter unit for transmitting the data over IR and a receiver for receiving the data.
  - Infrared Light Emitting Diode (LED) is the IR source for transmitter and at the receiving end a photodiode acts as the receiver
  - Both transmitter and receiver unit will be present in each device supporting IR communication for bidirectional data transfer
  - IR units are known as 'Transceiver'. Certain devices like a TV remote control always require unidirectional communication and so they contain either the transmitter or receiver unit (The remote control unit contains the transmitter unit and TV contains the receiver unit)
  - 'Infra-red Data Association' is the regulatory body responsible for defining and licensing the specifications for IR data communication



Basic IR Communication

How the subsystem in ES ………… ?

- Infrared(IR)……..
  - IrDA communication has two essential parts; a physical link part and a protocol part.
    - The physical link is responsible for the physical transmission of data between devices supporting IR communication and
    - protocol part is responsible for defining the rules of communication
  - The physical link works on the wireless principle making use of Infrared for communication.
  - The IrDA specifications include the standard for both physical link and protocol layer.
  - The IrDA control protocol contains implementations for Physical Layer (PHY), Media Access Control (MAC) and Logical Link Control (LLC)
  - The Physical Layer defines the physical characteristics of communication like range, data rates, power, etc
  - IrDA is a popular interface for file exchange and data transfer in low cost devices.
  - IrDA was the prominent communication channel in mobile phones before Bluetooth's existence. Even now most of the mobile phone devices support IrDA

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How the subsystem in ES ………… ?

- Bluetooth…..
  - is a low cost, low power, short range wireless technology for data and audio communication.
  - is a wireless technology standard for short distances (using short-wavelength UHF band from 2.4 to 2.485 GHz)for exchanging data over radio waves in the ISM and mobile devices, and building personal area networks (PANs)
  - Uses the Frequency Hopping Spread Spectrum (FHSS) technique for communication
    - Bluetooth divides transmitted data into packets, and transmits each packet on one of 79 designated Bluetooth channels
    - Each channel has a bandwidth of 1 MHz. It usually performs 800 hops per second, with Adaptive Frequency-Hopping (AFH) enabled
  - Invented by telecom vendor Ericsson in 1994, it was originally conceived as a wireless alternative to RS- 232 data cables
  - IR, Bluetooth communication also has two essential parts; a physical link part and a protocol part
  - The physical link is responsible for the physical transmission of data between devices supporting Bluetooth communication and protocol part is responsible for defining the rules of communication
  - The physical link works on the Wireless principle making use of RF(Radio Frequency) waves for communication
  - It supports a data rate of up to 1Mbps to 50Mbps (and a range of approximately30 to 100 feet (Depending on the Bluetooth version)

How the subsystem in ES ……….. ?

- Bluetooth….

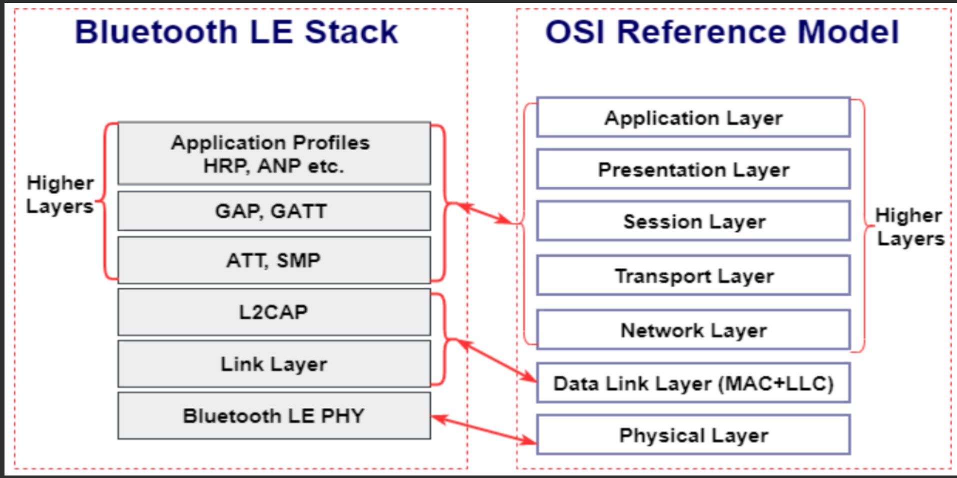
Version	Release Year	Max Transmission Speed	Max Range	Key Features / Improvements	Typical Use Cases
1.0 — 1.2	1999—2003	721 kbps	10 m (33 ft)	Basic wireless connections, Adaptive Frequency Hopping (AFH)	Early headsets, basic data transfer
2.0 — 2.1	2004—2007	3 Mbps (EDR)	10 m (33 ft)	Enhanced Data Rate (EDR), Secure Simple Pairing (SSP)	Phones, hands-free kits
3.0 + HS	2009	24 Mbps (with Wi-Fi HS)	10 m (33 ft)	High-Speed (HS) mode using Wi-Fi assist	Large file transfer
4	2009	1 Mbps (LE), 3 Mbps (EDR)	60 m (200 ft)	Introduction of Bluetooth Low Energy (BLE)	Wearables, fitness trackers
4.1	2013	1 Mbps (LE), 3 Mbps (EDR)	60 m (200 ft)	Better coexistence with LTE, device roles switching	Smart devices
4.2	2014	1 Mbps (LE), 3 Mbps (EDR)	60 m (200 ft)	Improved privacy, IPv6/6LoWPAN support	IoT sensors, healthcare devices
5	2016	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	2× speed, 4× range, larger broadcast capacity	Smart home, PC peripherals
5.1	2019	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	Direction finding (cm-level accuracy)	Asset tracking, indoor navigation
5.2	2020	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	LE Audio, LC3 codec, Enhanced Attribute Protocol (EATT)	Wireless earbuds, conferencing
5.3	2021	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	Reduced power consumption, more stable connections	Multi-device setups
5.4	2023	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	Periodic Advertising with Responses (PAwR), Encrypted Advertising Data (EAD)	Smart home networks, IoT mesh devices
6	2024	2 Mbps (LE), 50 Mbps (EDR)	240 m (800 ft)	Channel Sounding for centimeter-level precision, enhanced IoT features	Digital keys, smart cities, advanced IoT

Bluetooth Version Comparison Table

How the subsystem in ES ………… ?

- Bluetooth…..
  - The rules governing the Bluetooth communication is implemented in the 'Bluetooth protocol stack'.
  - The Bluetooth communication IC holds the stack. Each Bluetooth device will have a 48 bit unique identification number.
  - Bluetooth communication follows packet based data transfer. Bluetooth supports point-to-point (device to device) and point-to-multipoint (device to multiple device broadcasting) wireless communication.
  - The point to-point communication follows the master slave relationship. A Bluetooth device can function as either master or slave.
  - A network is formed with one Bluetooth device as master and more than one device as slaves, it is called a Piconet. A Piconet supports a maximum of seven slave devices
  - Bluetooth is the favourite choice for short range data communication in handheld embedded devices.
  - Bluetooth technology is very popular among cell phone users as they are the easiest communication channel for transferring ringtones, music fi les, pictures, media fi les, etc. between neighboring Bluetooth enabled phones.
  - Bluetooth Low Energy (BLE)/Bluetooth Smart is a latest addition to the Bluetooth technology. BLE allows devices to use much less power compared to the standard Bluetooth connections, while offering most of the connectivity of Bluetooth and maintaining a similar communication range

How the subsystem in ES ………… ?

- Bluetooth…..
  - The Bluetooth standard specifies the minimum requirements that a Bluetooth device must support for a specific usage scenario
  - The Generic Access Profile (GAP) defines the requirements for detecting a Bluetooth device and establishing a connection with it
  - All other specific usage profiles are based on GAP.
    - Serial Port Profile (SPP) for serial data communication,
    - File Transfer Profile (FTP) for file transfer between devices,
    - Human Interface Device (HID) for supporting human interface devices like keyboard and mouse are examples for Bluetooth profiles
  - BLE implements various application specific profiles for communicating with low power Bluetooth peripherals like fitness devices, Blood Pressure and heart rate monitors etc. Healthcare Profiles (HTP, GLP, BLP etc.), Sports and Fitness Profiles (HRP, LNP, RNCP etc.)
  - 

[Bluetooth LE protocol stack is shown along with the OSI reference model]()

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How the subsystem in ES ………… ?

- Wi-Fi…..
  - Wi-Fi or Wireless Fidelity is the popular wireless communication technique for networked communication of devices
  - Wi-Fi follows the IEEE 802.11 standard.
  - Wi-Fi is intended for network communication and it supports Internet Protocol (IP) based communication
  - Wi-Fi based communications require an intermediate agent called Wi-Fi router/Wireless Access point to manage the communications
  - The Wi-Fi router is responsible for restricting the access to a network, assigning IP address to devices on the network, routing data packets to the intended devices on the network
  - Wi-Fi enabled devices contain a wireless adaptor for transmitting and receiving data in the form of radio signals through an antenna , The hardware part of it is known as Wi-Fi Radio.
  - Wi-Fi operates at 2.4GHz or 5GHz of radio spectrum and they co-exist with other ISM band devices like Bluetooth
  - For communicating with devices over a Wi-Fi network, the device when its Wi-Fi radio is turned ON, searches the available WiFi network in its vicinity and lists out the Service Set Identifier (SSID) of the available networks

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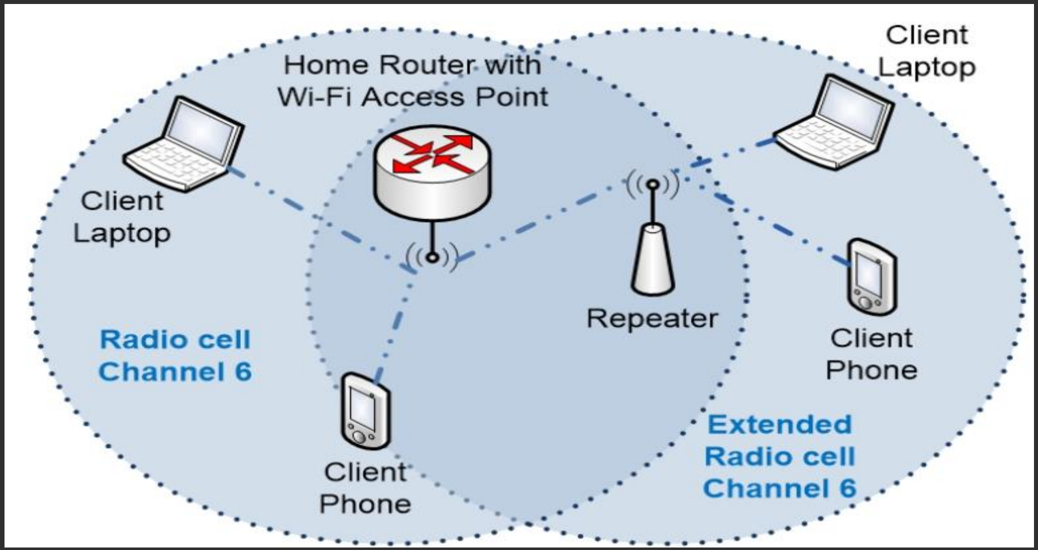
How the subsystem in ES ………… ?

- Wi-Fi…..
  - For the security enabled network, a password may be required to connect to a particular SSID.
  - Wi-Fi employs different security mechanisms like Wired Equivalency Privacy (WEP) Wireless Protected Access (WPA) etc.. for securing the data communication
  - Wi-Fi supports data rates ranging from 1Mbps to 1300Mbps (Growing towards higher rates as technology progresses),
  - Depending on the standards (802.11a/b/g/n/ac) and access/ modulation method.
  - Depending on the type of antenna and usage location (indoor/outdoor),Wi-Fi offers a range of 100 to 1000 feet



Common Wi-Fi network solutions

How the subsystem in ES ………… ?

- ZigBee
  - ZigBee is a low power, low cost, wireless network communication protocol based on the IEEE 802.15.4-2006 standard
  - ZigBee is targeted for low power, low data rate and secure applications for Wireless Personal Area Networking (WPAN)
  - The ZigBee specifications support a robust mesh network containing multiple nodes.
  - This networking strategy makes the network reliable by permitting messages to travel through a number of different paths to get from one node to another
  - ZigBee operates worldwide at the unlicensed bands of Radio spectrum, mainly at:
    - 2.400 to 2.484 GHz,
    - 902 to 928 MHz
    - and 868.0 to 868.6 MHz.
  - ZigBee Supports an operating distance of up to 100 metres and a data rate of 20 to 250Kbps

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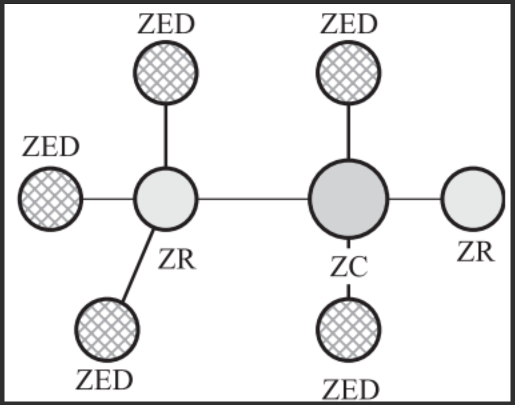
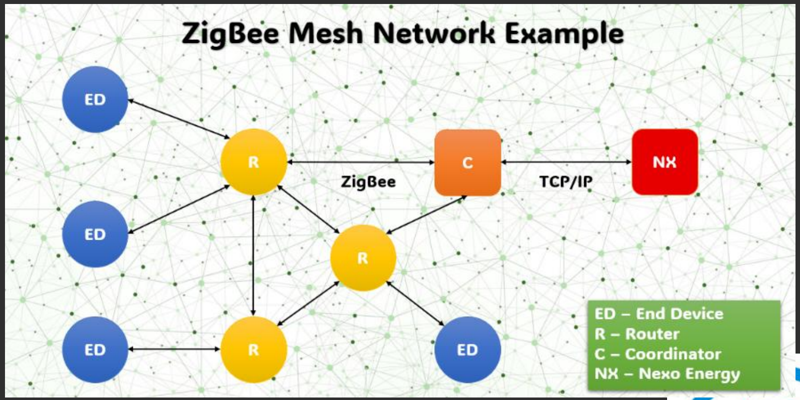
How the subsystem in ES ………… ?

- ZigBee…..
  - ZigBee terminology, each ZigBee device falls under any one of the following ZigBee device category :
    - ZigBee Coordinator (ZC)/Network Coordinator
      - The ZigBee coordinator acts as the root of the ZigBee network.
      - The ZC is responsible for initiating the ZigBee network and it has the capability to store information about the network.
    - ZigBee Router (ZR)/Full function Device (FFD)
      - Responsible for passing information from device to another device or to another ZR
    - ZigBee End Device (ZED)/Reduced Func on Device (RFD)
      - End device containing ZigBee functionality for data communication
      - It can talk only with a ZR or ZC and doesn't have the capability to act as a mediator for transferring data from one device to another



A ZigBee network model



[ZigBee communication protocol]

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How the subsystem in ES ………… ?

- ZigBee…..
  - ZigBee is primarily targeting application areas like
    - Home & industrial automation,
    - Energy management,
    - Home control/security,
    - Medical/patient tracking,
    - Logistics & asset tracking and sensor networks & active RFID
    - Automatic Meter Reading (AMR),
    - Smoke detectors,
    - Wireless telemetry,
    - HVAC control,
    - Heating control,
    - lighting controls, environmental controls, etc.,

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How the subsystem in ES ………… ?

- ZigBee…..
  - ZigBee PRO offers full wireless mesh, low-power networking capable of supporting more than 64,000 devices on a single network
  - It provides standardised networking designed to connect the widest range of devices, in any industry, into a single control network
  - ZigBee PRO offers an optional new and innovative feature, ‘Green Power’ to connect energy harvesting or self-powered devices into ZigBee PRO networks.
  - The ‘Green Power’ feature of ZigBee PRO is the most eco-friendly way to power battery-less devices such as sensors, switches, dimmers and many other devices and allows them to securely join ZigBee PRO networks
  - ZigBee 3.0 delivers all the features of ZigBee while unifying the ZigBee application standards found in ZigBee devices.
  - ZigBee 3.0 standard enables communication and interoperability among devices for smart homes, connected lighting, and other market

  The specifications for ZigBee is developed and managed by the ZigBee alliance

How the subsystem in ES ………… ?

- General Packet Radio Service (GPRS), 3G, 4G, LTE
  - General Packet Radio Service (GPRS), 3G, 4G,5G and LTE are cellular communication technique for transferring data over a mobile communication network like GSM and CDMA
  - Data is sent as packets in GPRS communication.
  - The transmitting device splits the data into several related packets.
  - At the receiving end the data is re-constructed by combining the received data packets.
  - GPRS supports a theoretical maximum transfer rate of 171.2kbps
  - In GPRS communication, the radio channel is concurrently shared between several users instead of dedicating a radio channel to a cell phone user
  - The GPRS communication divides the channel into 8 timeslots and transmits data over the available channel.
  - GPRS supports Internet Protocol (IP), Point to Point Protocol (PPP) and X.25 protocols for communication.
  - GPRS is mainly used by mobile enabled embedded devices for data communication.
  - The device should support the necessary GPRS hardware like GPRS modem and GPRS radio
  - To accomplish GPRS based communication, the carrier network also should have support for GPRS communication
  - To accomplish GPRS based communication, the carrier network also should have support for GPRS communication

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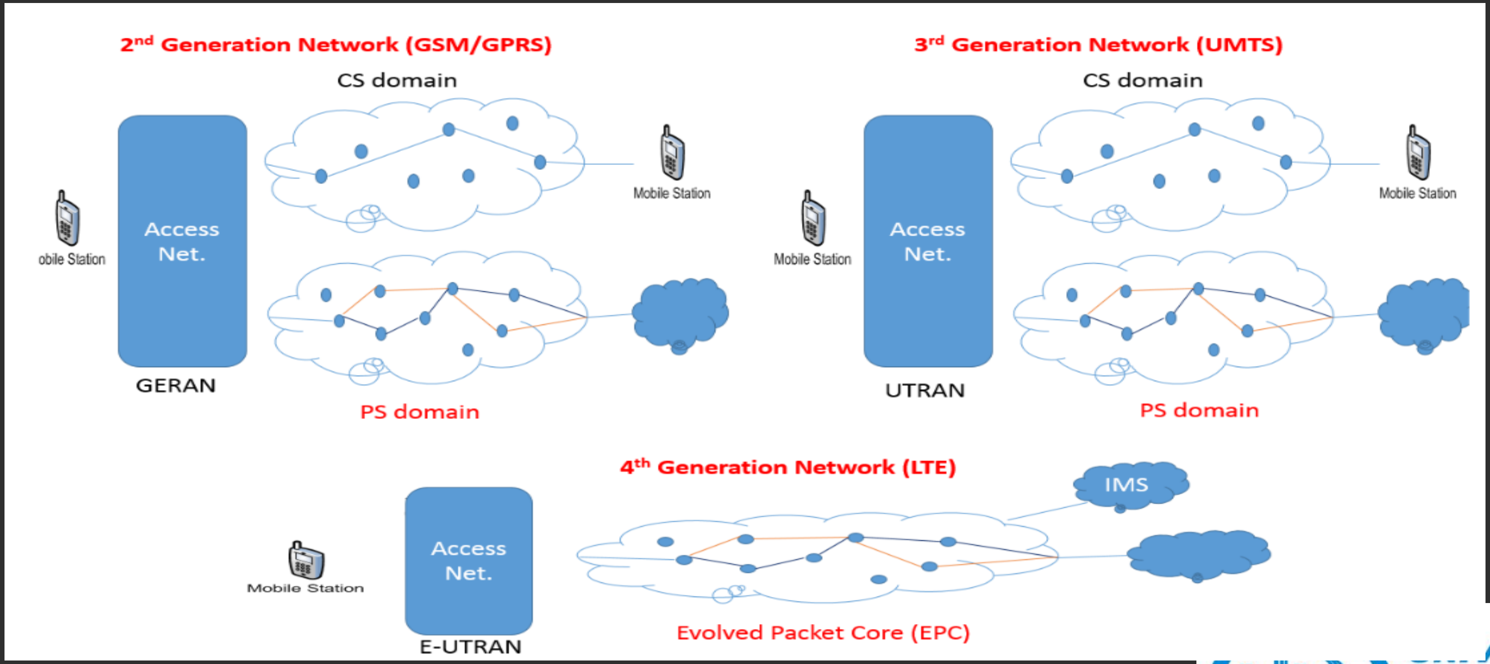
How the subsystem in ES ………… ?

- General Packet Radio Service (GPRS), 3G, 4G, LTE
  - GPRS is an old technology and it is being replaced by new generation cellular data communication techniques like 3G (3rd Generation), High Speed Downlink Packet Access (HSDPA), 4G (4th Generation), LTE (LongTerm Evolution) etc.
  - Offers higher bandwidths for communication.
  - 3G offers data rates ranging from 144Kbps to 2Mbps or higher,
  - 4G gives a practical data throughput of 2 to 100+ Mbps depending on the network and underlying technology
  - 5G gives a practical data throughput to 200+ Mbps depending on the network



Cellular communication

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How the subsystem in ES ………… ?

- Mobile network throughput comparison, 3G, 4G, 5G ,LTE

Comparison	2G	3G	4G	5G
Introduced in year	1993	2001	2009	2018
Technology	GSM	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	TDMA, CDMA	CDMA	CDMA	OFDM, BDMA
Switching type	Circuit switching for voice and packet switching for data	Packet switching except for air interference	Packet switching	Packet switching
Internet service	Narrowband	Broadband	Ultra broadband	Wireless World Wide Web
Bandwidth	25 MHz	25 MHz	100 MHz	30 GHz to 300 GHz
Advantage	Multimedia features (SMS, MMS), internet access and SIM introduced	High security, international roaming	Speed, high speed handoffs, global mobility	Extremely high speeds, low latency
Applications	Voice calls, short messages	Video conferencing, mobile TV, GPS	High speed applications, mobile TV, wearable devices	High resolution video streaming, remote control of vehicles, robots, and medical procedures

[Comparison of 2G, 3G, 4G, and 5G]

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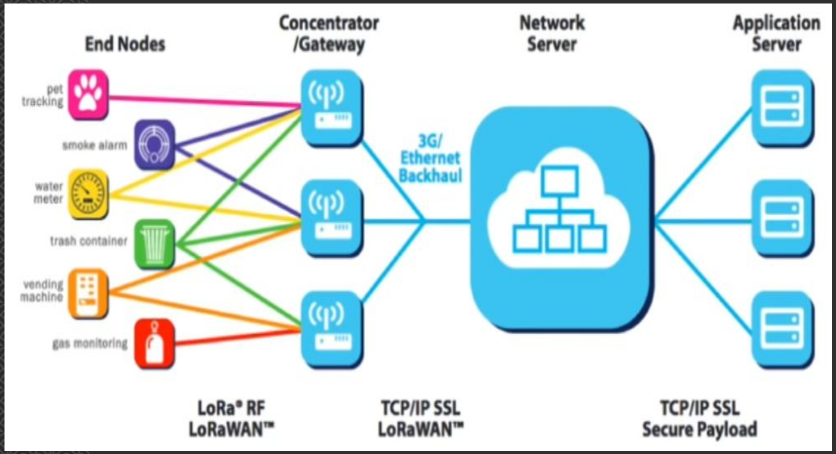
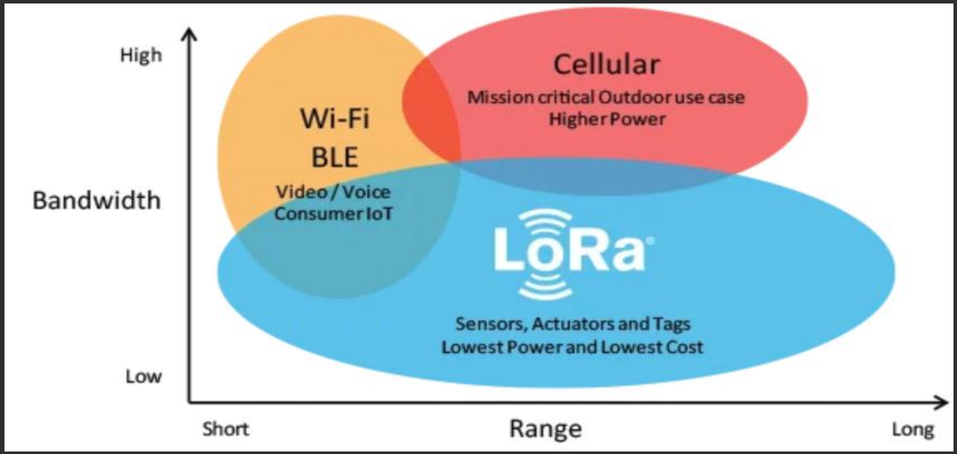
How the subsystem in ES ………… ?

- LoRa WAN
  - stands for Low Power Wide Area Network and this type of wireless communication is designed for sending small data packages over long distances, operating on a battery
  - Data rates Between 0.3 kbps to 5.5 kbps
  - Gateways can handle 100s of devices at the same time.
  - The gateways can listen to multiple frequencies simultaneously, in every spreading factor at each frequency.
  - Communications are bidirectional



[LoRa]



[LoRa and LoRaWAN]

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How the subsystem in ES ………… ?

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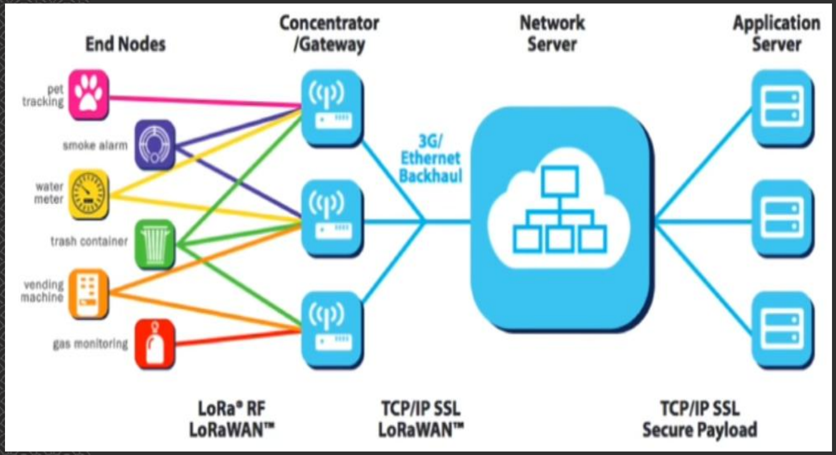
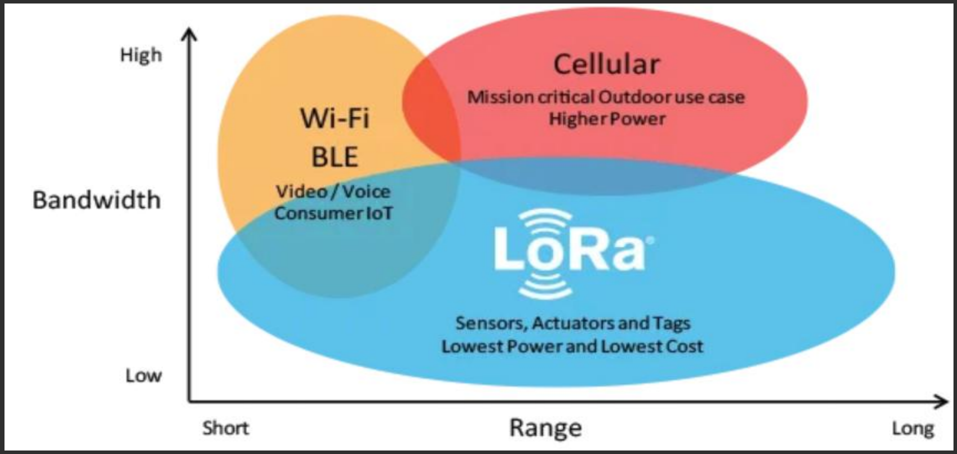


LoRa



LoRa and LoRaWAN

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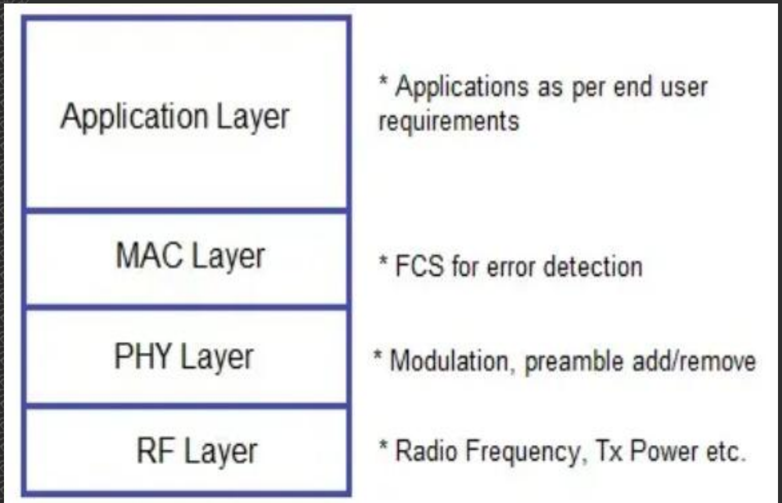
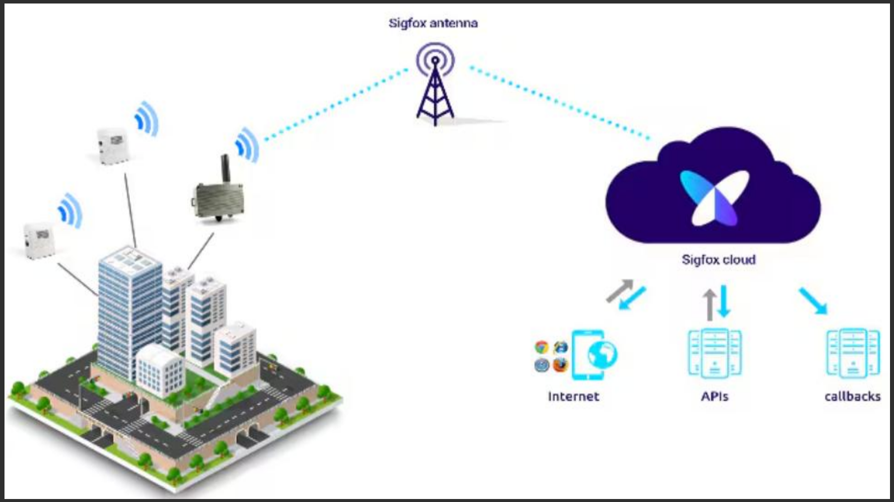
How the subsystem in ES ………… ?

- SigFox
  - The SigFox protocol stack is designed for efficient, low-power communication within its Low Throughput Network (LTN).
  - It's built with several layers, each handling specific tasks like physical signal transmission, data framing, and message processing.
  - This guide breaks down the architecture of the SigFox protocol stack and explains how each layer contributes to smooth IoT communication



A simplified protocol stack of the SigFox wireless system



Sigfox

Assignment

- Make slides for each group
  - Onboard
  - Group1

1.Inter-Integrated Circuit (I2C or I2C) Bus
2. Serial Peripheral Interface (SPI) Bus
3.UART

Grroup2

3. Controlled Area Network (CAN)
4. Parallel Interface

- For **External Communication Interfaces:**
  - Group3

1. RS-232C & RS-485
2. Universal Serial Bus (USB)

Group4

3. Infrared
4. Bluetooth
5. Wi-Fi

Group5

6. ZigBee
7.Lora
8. Sigfox
9 GSM
8. General Packet Radio Service (GPRS)

Deadline 11pm 16/04/2026

References

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- What Is SPI Protocol, SPI Tutorial With Code, Working, Applications
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- Raspberry Pi Serial Communication: What, Why, and a Touch of How
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- An Introduction to CAN Bus
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- UART, RS232, RS485, I2C, SPI Protocols and Realization
- RS-485 Wiring, Communication and Difference with RS232
- RS232 vs RS485 – What are their differences?
- A Complete Analysis of the USB Protocol
- Introduction To USB Protocol
- IrDA Communications Protocols
- IrDA infrared wireless communications: protocol throughput optimization
- Bluetooth Architecture from Scratch
- Bluetooth® market dashboard
- PC to PC communication over Zigbee Protocol using Xbee and XCTU : IoT Part 36
- What Is Zigbee Wireless Technology And How It Works?
- Mobile network
- Comparing LoRaWan and Sigfox is like Comparing apples and oranges